

Pathogenesis and Risk of Cardiovascular Disease in Chronic Kidney Disease

Peter Stenvinkel, Nikolaus Marx, Charles A. Herzog

Diminished estimated glomerular filtration rate (eGFR) and albuminuria are powerful graded, independent predictors of cardiovascular (CV) morbidity and mortality¹ and all-cause mortality. Even subtle kidney dysfunction, as suggested by albuminuria, increases CV risk because it may reflect microvasculature health, including endothelial function. Patients with end-stage kidney disease (ESKD) face an extraordinary risk for premature death, largely because of CV complications in the setting of accelerated vascular aging. Patients with eGFR less than 60 mL/min/1.73 m² are much more likely to die than to develop ESKD, reflecting the burden of cardiovascular disease (CVD) in this population. The most effective strategy for reducing CV morbidity and mortality in this high-risk population would be to target patients with mildly reduced eGFR for prevention and treatment.

Patients with chronic kidney disease (CKD) were often excluded from randomized controlled trials targeting CVD, possibly reducing acceptance of evidence-based therapies (validated in nonrenal patients) and fostering “therapeutic nihilism” in clinicians who treat CKD patients. Thus, novel treatment strategies are urgently needed to reduce the unacceptable high CV event rate.

Like conventional atheromatous occlusive vascular disease, CKD is characterized by generalized vasculopathy, with other characteristics, including left ventricular hypertrophy, vascular calcification, and vascular noncompliance. Numerous CVD risk factors are specific to CKD and operate in addition to conventional risk factors found in the general population.

EPIDEMIOLOGY

Prevalence of Cardiovascular Complications in Chronic Kidney Disease

Interpretation of epidemiologic studies of CVD is problematic because of the difficulty in defining cause of death. Unexpected sudden death most likely results from arrhythmia, but a subarachnoid hemorrhage, massive embolic stroke, or aortic dissection might be indistinguishable from a primary arrhythmic event without an autopsy. Defining “coronary heart disease” (CHD) is also problematic: in the general population, sudden cardiac death is a primary complication of CHD, but this is unlikely to be true for dialysis patients. A history of angina cannot reliably classify a patient as having CHD because angina (resulting from supply-demand mismatch) can occur in patients with left ventricular hypertrophy and angiographically pristine coronary arteries. This probably relates to the increased myocardial fibrosis, diminished relative capillary density, and increased thickening of the intramyocardial vessel walls in uremia. At lower levels of eGFR (especially in dialysis patients), the burden of nonatherosclerotic (vs. atherosclerotic) CVD is relatively increased (Fig. 84.1).² Although occlusive CHD is common in CKD, acute myocardial infarction (MI) and coronary artery disease account

for only 8% of cardiac deaths; 85% of cardiac deaths of hemodialysis (HD) patients in the US Renal Data System (USRDS) database are attributable to arrhythmic mechanisms (www.USRDS.org), continuing a temporal trend of an increasing proportion of deaths attributable to arrhythmic mechanisms. Data from the ERA-EDTA registry are encouraging because the absolute excess mortality for atheromatous CVD decreased between 2007 to 2015.³

Of incident dialysis patients, at least 75% have left ventricular hypertrophy, and 75% to 85% have hypertension. Hypertension, anemia, vascular noncompliance, elevated fibroblastic growth factor 23 (FGF23) levels, and volume overload contribute to left ventricular hypertrophy. Based on echocardiography, 85% to 90% of patients have a left ventricular ejection fraction of 50% or greater despite frequent congestive heart failure (i.e., heart failure with preserved ejection fraction [HFpEF]). Therefore, many volume overload episodes in dialysis patients may be attributable to diastolic dysfunction or circulatory congestion. Fig. 84.2 provides a snapshot of CVD event rates (by modality) in prevalent ESKD patients in 2018.⁴

Cardiovascular Disease Is Present Long Before the Start of Kidney Replacement Therapy

In elderly patients with stage 2 to 3 CKD, traditional risk factors seem to be the major contributors to CV mortality. Atherosclerosis Risk in Communities (ARIC) data suggest that both traditional and novel risk factors are relevant at CKD stage 4, and novel risk factors are far more prevalent in dialysis patients than in the general population (Fig. 84.3).⁵ The Framingham predictive instrument does not accurately predict coronary events in CKD. Mild to moderate CKD is associated with increased risk for venous thromboembolism, supporting the concept of hypercoagulability in CKD (see also Chapter 87). Because the incidence of CV events is much higher in the first weeks after HD initiation, concerns have been raised that the dialysis procedure per se may trigger CV events.⁶

Racial and International Differences in Cardiovascular Disease Prevalence

In the United States, survival is better for African American than for White dialysis patients. However, overall CV mortality among dialysis patients from the United States is significantly greater than is observed in Japan and Europe, even after adjustment for standard risk factors and dialysis dose. Higher mortality rates in US dialysis patients may be related to higher prevalence of sicker or diabetic patients, differences in dialysis practice patterns, cultural habits, differences in diet, or genetic variations.

Reverse Epidemiology

Reverse epidemiology (preferably called *confounded epidemiology*) refers to the paradoxical observation that the association among hypercholesterolemia, hypertension, obesity, and poor outcomes, including CV

Change in Cardiovascular Risk During CKD Progression

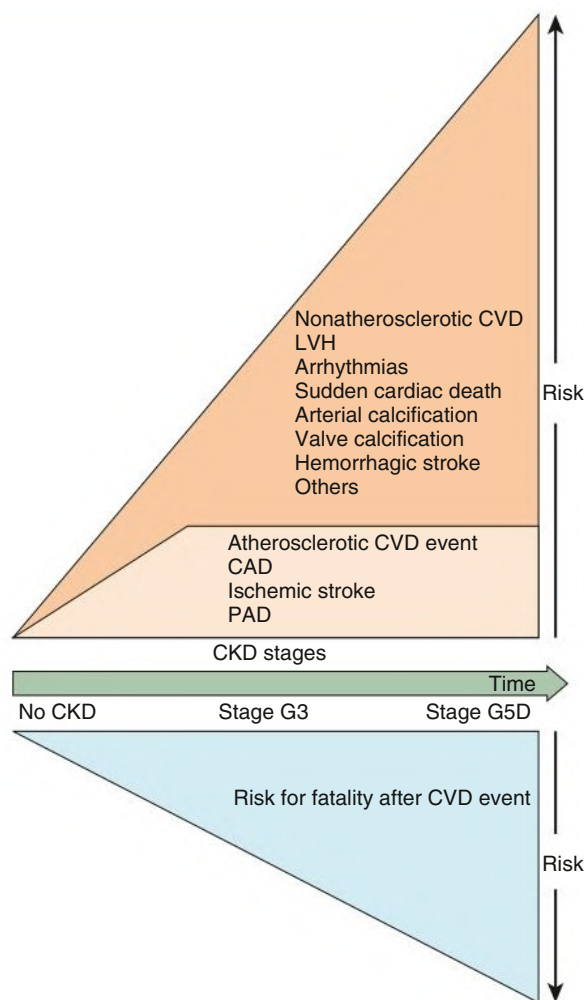


Fig. 84.1 Change in Cardiovascular Risk During Chronic Kidney Disease (CKD) Progression. Cardiovascular disease (CVD) event (upper triangle), contributions of atherosclerotic CVD (tan), nonatherosclerotic CVD (orange), and risk for fatality after CVD event (blue). CAD, coronary artery disease; LVH, left ventricular hypertrophy; PAD, Peripheral artery disease. (From Wanner C, Amann K, Shoji T. The heart and vascular system in dialysis. *Lancet*. 2016;388:76–284.)

death, in the general population does not exist and may be reversed in CKD. Patients with wasting and inflammation appear to mostly account for poor survival and confounded epidemiology.

ETIOLOGY AND RISK FACTORS

Traditional Risk Factors

Age, Sex, and Smoking

The US National Health and Nutrition Examination Surveys (NHANES) shows the prevalence of CV factors and CVD prevalence in relation to age and CKD stage. In the United States, the average age in 2018 at kidney replacement therapy (KRT) initiation was 63 years,⁴ when CVD is common in the general population. The mean survival for patients who start dialysis in the United States is 3 years. An individual-level meta-analysis including more than 2 million participants showed that low eGFR and high albuminuria were independently associated with mortality and ESKD regardless of age. Female sex is associated with a 4% independent increased risk for mortality in incident dialysis

patients and smoking with a 52% increased risk for death in dialysis patients.

Diabetes Mellitus

Diabetes accounted for 47% of incident US ESKD patients between 2014 and 2018⁴ and is a common cause of ESKD in many countries. Diabetic patients starting KRT have numerous cardiovascular risk factors, including dyslipidemia, hypertension, persistent inflammation, gut dysbiosis, increased oxidative stress, and protein-energy wasting. Diabetes at dialysis initiation is a potent independent risk factor for all-cause and CVD-related deaths, including after coronary revascularization or acute MI. Nevertheless, the rate of incident acute MI is even higher for patients with CKD stages 3b-4 without diabetes than for patients with diabetes and CKD stages 1 to 2.

Hypertension

Hypertension predicts mortality in CKD patients before or at dialysis initiation. Isolated systolic hypertension with increased pulse pressure

Overall Prevalence of Common Cardiovascular Diseases in Adult Patients With End-Stage Kidney Disease

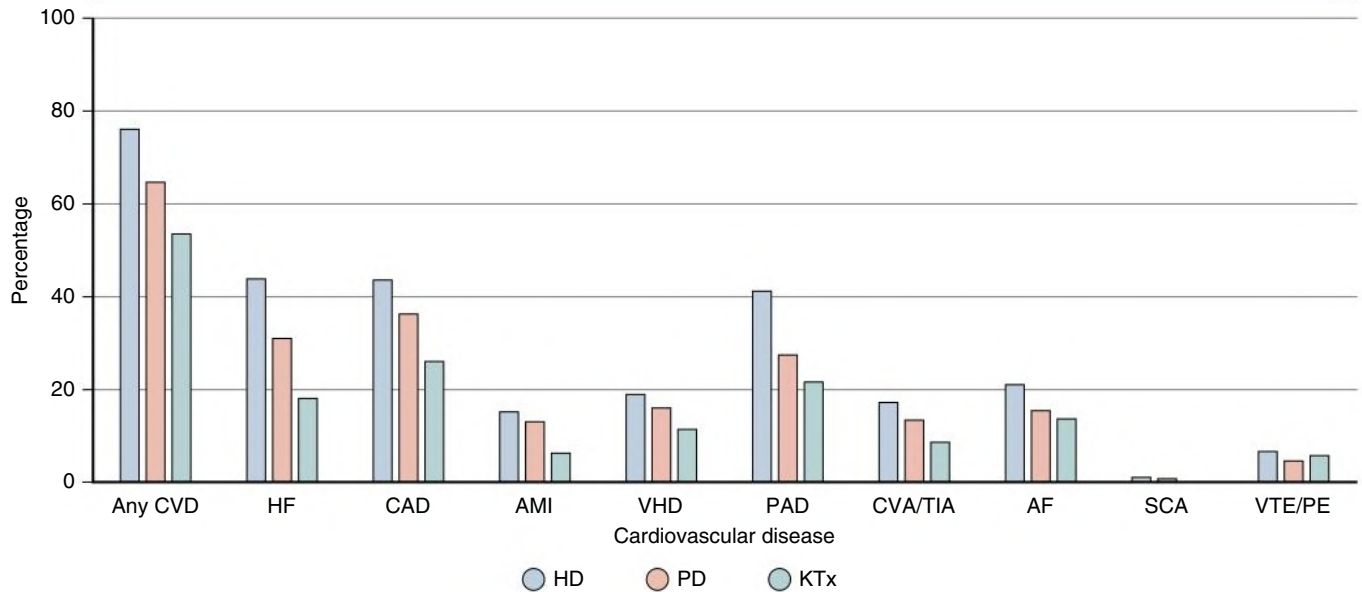


Fig. 84.2 Unadjusted Prevalence of Common Cardiovascular Diseases in Adult Patients with End-stage Kidney Disease, by Treatment Modality, 2018. AF, Atrial fibrillation; AMI, acute myocardial infarction; CAD, coronary artery disease; CVA, cerebrovascular accident; CVD, cardiovascular disease; HF, heart failure; KTx, kidney transplant; PAD, peripheral arterial disease; PD, peritoneal dialysis; PE, pulmonary embolism; SCA, sudden cardiac arrest; TIA, transient ischemic attack; VHD, valvular heart disease; VTE, venous thromboembolism. (From US Renal Data System. *USRDS 2020 Annual Data Report: Atlas of Chronic Kidney Disease & End-Stage Renal Disease in the United States*. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2020.)

Risk Factors for Cardiovascular Disease in Chronic Kidney Disease

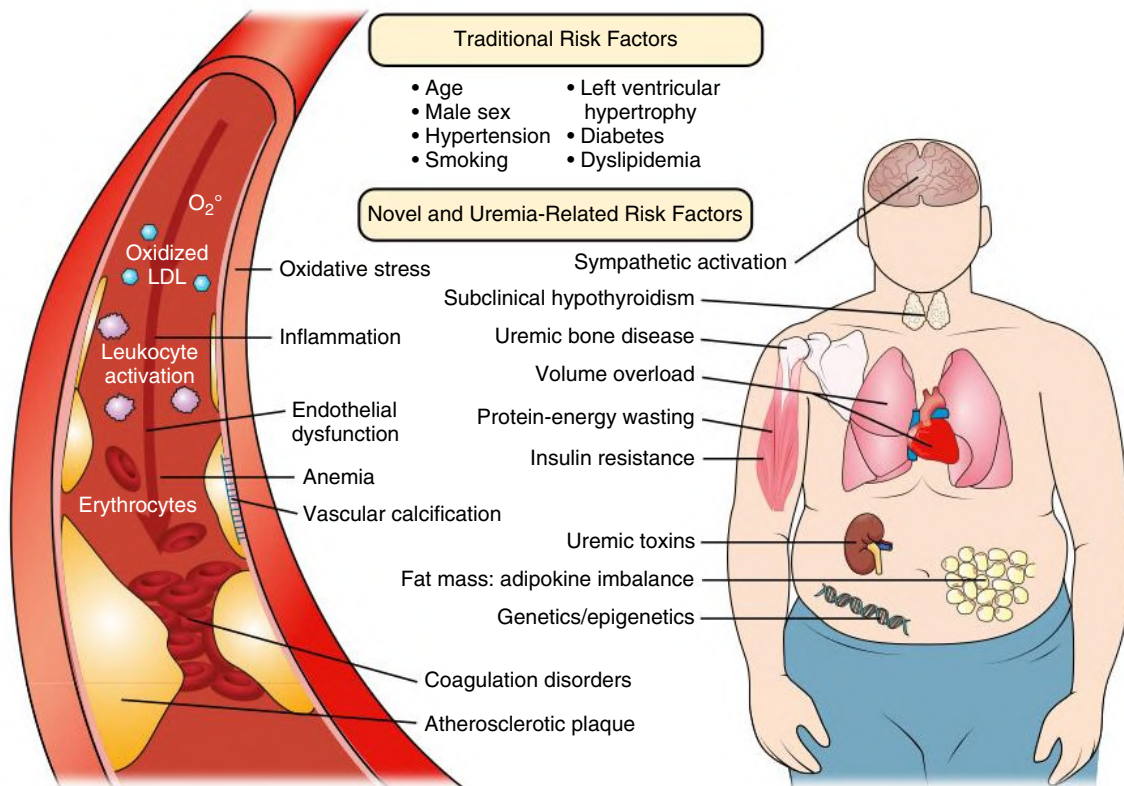


Fig. 84.3 Risk Factors for Cardiovascular Disease in Chronic Kidney Disease. Overview of traditional (i.e., Framingham) risk factors and novel and uremia-related risk factors. LDL, Low-density lipoprotein. (From Stenvinkel P, Carrero JJ, Axelsson J, Lindholm B, Heimbürger O, Massy Z. Emerging biomarkers for evaluating cardiovascular risk in the chronic kidney disease patient: how do new pieces fit into the uremic puzzle? *Clin J Am Soc Nephrol*. 2008;3:505–521.)

TABLE 84.1 Common Patterns of Hyperlipidemia in Different Stages of Kidney Disease, Compared With the Healthy Population

Stage of Kidney Disease	CHOLESTEROL			
	Total	High-Density Lipoproteins	Low-Density Lipoproteins	Triglycerides
Nephrotic syndrome	↑↑↑	↓	↑↑	↑
Chronic kidney disease	No change	↓	No change ^a	↑↑
Hemodialysis	No change	↓	No change ^a	↑↑
Peritoneal dialysis	↑	↓	↑	↑
Transplantation	↑↑	No change	↑	↑

^aComposition altered.

is by far the most prevalent blood pressure anomaly in dialysis patients, resulting from arterial medial sclerosis (i.e., arteriosclerosis) with secondary vascular stiffening. Stiff vessels cause increased pulse wave velocity, resulting in increased systolic blood pressure peak pressure by a prematurely reflected pulse wave, progressive left ventricular dysfunction, and finally congestive heart failure. At this stage, mean arterial and diastolic pressure may decrease. The relationship between blood pressure and mortality is U-shaped; isolated systolic hypertension and increased pulse pressure probably indicate high long-term risk in dialysis patients, whereas low mean and diastolic blood pressures predict early mortality. CKD patients are frequently “nondippers” and experience sleep apnea and sympathetic nervous system activation.

Dyslipidemia

The relationship between hypercholesterolemia, CVD, and mortality in CKD is weak because common uremic CV abnormalities, such as cardiomyopathy and arteriosclerosis (i.e., vascular calcification), are less dependent on dyslipidemia than on other factors. Low rather than high serum cholesterol level is associated with poor survival in HD patients, likely related to confounding by protein-energy wasting and inflammation.

Progressive CKD leads to changes in blood lipids typically associated with vascular disease, including decreased apolipoprotein A (apoA)-containing lipoproteins and increased apoB-containing lipoproteins (Table 84.1). Serum triglycerides are elevated in most ESKD patients, whereas total serum cholesterol is variable, depending on nutritional status and presence of inflammation. High-density lipoprotein (HDL) cholesterol is typically reduced, and low-density lipoprotein (LDL), intermediate-density lipoprotein, and very low-density lipoprotein cholesterol, as well as lipoprotein(a) levels, tend to be increased. Because of altered molecular composition, the antiinflammatory activity of HDL is lost in the uremic milieu. Compared with long-term HD patients, peritoneal dialysis (PD) patients more often have both hypercholesterolemia and hypertriglyceridemia. Both groups are characterized by low HDL and elevated oxidized LDL cholesterol levels; elevated lipoprotein(a) levels are associated with increased CVD mortality.

Insulin Resistance and Atherosclerosis

In the general population, impaired insulin-stimulated glucose disposal in muscle is often part of a metabolic syndrome that includes dyslipidemia, hypertension, endothelial dysfunction, and sympathetic overactivity. Many of these abnormalities are present in CKD. Although insulin resistance was found to be an independent predictor of CV mortality in dialysis patients, its magnitude of contribution in CKD mortality is uncertain. High urea levels have been linked to insulin resistance.⁷

Nontraditional and Uremia-Specific Risk Factors

Even mild CKD is an independent risk factor for CVD and similar in magnitude to diabetes and hypertension. The uremic milieu may affect

both quality and quantity of the atherosclerotic plaques. Coronary lesions in uremic patients, compared with nonrenal controls, are characterized by increased media thickness, infiltration and activation of macrophages, and marked media calcification. The mechanism(s) by which a uremic milieu may promote vascular senescence and accelerate vascular aging⁸ are not well established, but prevalence and magnitude of several nontraditional risk factors, such as oxidative stress, inflammation, gut dysbiosis, and advanced glycation end-products (AGEs), increase as kidney function deteriorates and uremic toxins accumulate. Many uremic retention solutes, such as asymmetric dimethylarginine (ADMA), guanidine, indoxyl sulfate, trimethylamine N-oxide (TMAO), interleukin (IL)-6, and *p*-cresol, which accumulate in CKD, may have proatherogenic properties.⁹ Finally, failing kidneys produce fewer substances that may inhibit CVD and atherogenesis (e.g., renase, a soluble monoamine oxidase that regulates cardiac function and BP).

Oxidative Stress

Oxidative stress may be implicated in the pathogenesis of atherosclerosis, the increased risk for atherosclerotic CV events, protein-energy wasting, and anemia.¹⁰ Increased production of reactive oxygen species in the vascular wall characterizes atherosclerosis. Moderate CKD, and in particular ESKD, is a prooxidant state resulting from reduced antioxidant systems (vitamin C and selenium deficiency, reduced intracellular vitamin E levels, reduced activity of the cytoprotective transcription factor Nrf2, reduced glutathione system activity), and increased prooxidant activity is associated with advanced age, diabetes, chronic inflammation, retained uremic solutes, and dialysis membranes and solutions. Four oxidative stress pathways operate in CKD: carbonyl stress, nitrosative stress, chlorinated stress, and classic oxidative stress.

Inflammation

Most dialysis patients are in a state of chronic low-grade inflammation, which associate with premature aging processes (i.e., inflammaging).¹¹ Inflammatory biomarkers, such as C-reactive protein (CRP), interleukin-6 (IL-6), pentraxin 3 (PTX3), fibrinogen, and white blood cell count, are independent predictors of mortality in CKD. It has been argued that repeated CRP measurements will improve dialysis patient care because persistent inflammation (i.e., CRP > 5 mg/L) may be a silent reflection of different pathophysiological alterations that require attention.¹² Hypoalbuminemia, strongly associated with systemic inflammation, is another strong outcome predictor in CKD. Whereas both dialysis-related factors and non-dialysis-related factors, infection, gut dysbiosis with a leaky gut, comorbidity, genetic factors, hypogonadism, diet, and kidney function loss may contribute to chronic inflammation, its primary causes are not always evident. The senescence-associated secretory phenotype caused by increased numbers of senescent cells in the uremic milieu¹³ also may be a

Effect of Altered Cytokine Production in Uremia on Various Target Organs

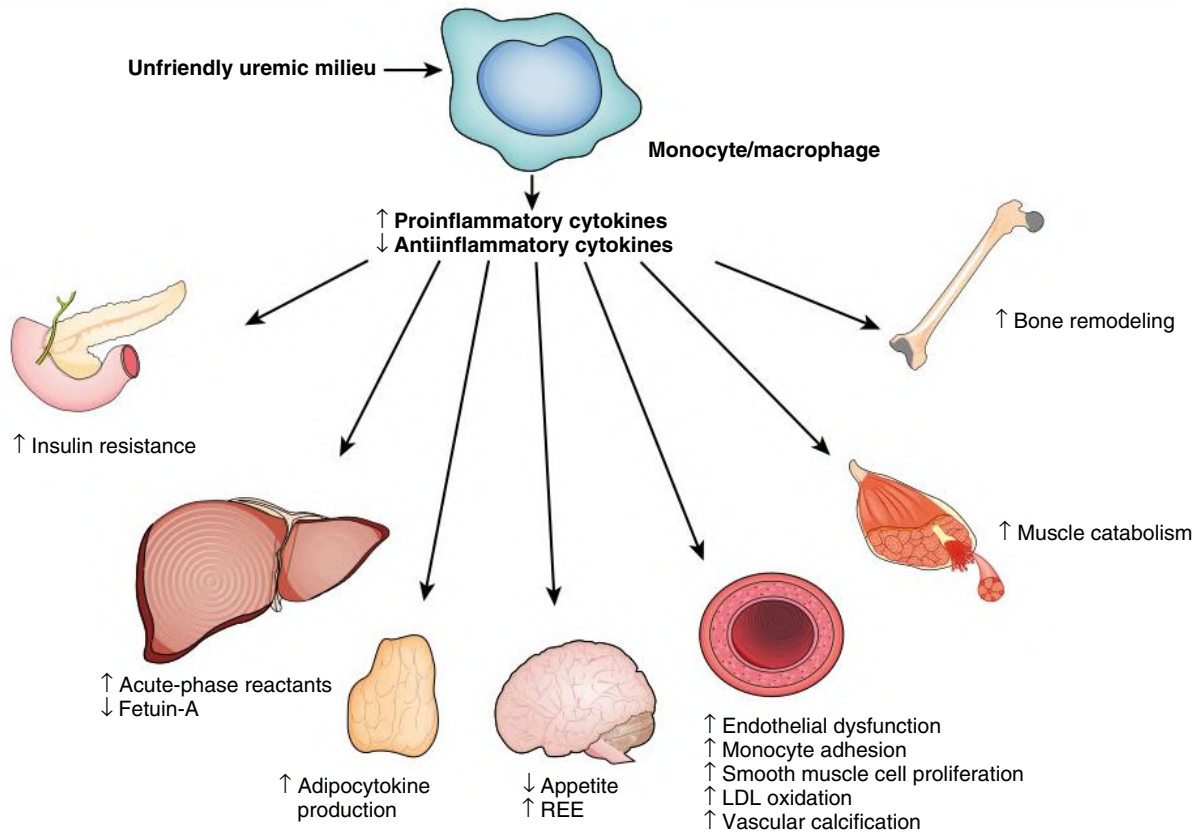


Fig. 84.4 Potential mechanisms by which elevated circulating levels of proinflammatory and anti-inflammatory cytokines may promote accelerated atherosclerosis, other uremic complications, and wasting. LDL, Low-density lipoprotein; REE, resting energy expenditure. (From Stenvinkel P, Carrero JJ, Axelsson J, Lindholm B, Heimbürger O, Massy Z. Emerging biomarkers for evaluating cardiovascular risk in the chronic kidney disease patient: how do new pieces fit into the uremic puzzle? *Clin J Am Soc Nephrol.* 2008;3:505–521.)

significant contributor to uremic inflammation. Whether the acute-phase response reflects only established atherosclerotic disease or is involved in the initiation and progression of atherosclerosis is unclear. Some inflammatory biomarkers, such as IL-6, PTX3, and tumor necrosis factor, have proatherogenic properties, such as promoting vascular calcification, oxidative stress, and endothelial dysfunction (Fig. 84.4). Evidence suggests associations between inflammation and development of albuminuria. Persistent inflammation may change the risk factor profile for traditional risk factors, such as cholesterol and obesity.¹⁴ The link between septicemia and subsequent increased risk for death and CV events, including acute MI, further supports inflammation as a trigger for CV events.

Gut Dysbiosis

The uremic milieu affects the intestinal microbiota and the integrity of the intestinal wall, possibly contributing to inflammation and increased production of uremic toxins, such as proatherogenic indoxyl sulfate, phenylacetic acid, and *p*-cresol sulfate. The uremic milieu and volume retention may damage the intestinal wall, promoting increased leakiness of endotoxins and translocation of intestinal bacteria. The microbial metabolite and water-soluble uremic toxin TMAO contributes to the development of atherosclerotic heart disease and vascular

calcification, and it is related to adverse CV outcomes in both the general population and in CKD.¹⁵

Endothelial Dysfunction

Endothelial dysfunction (as evaluated by impaired endothelium-dependent vasodilation) is common in CKD. Reasons include inflammation, ADMA retention, elevated TMAO, oxidative stress, elevated FGF-23, dyslipidemia, hyperglycemia, and hypertension. Serum ADMA concentrations are associated with endothelial function in uremic resistance vasculature. Surrogate markers of endothelial dysfunction, such as ADMA, PTX3, and adhesion molecules, independently predict death.¹⁶ Detached circulating endothelial cells are potential markers of endothelial damage and have prognostic value in HD patients. Normally, in response to an ischemic insult and cytokine stimulation, endothelial progenitor cells are mobilized from the bone marrow to repair endothelial injury, and this seems to be impaired in CKD.

Anemia

Anemia is a major cause of left ventricular hypertrophy and left ventricular dilation in CKD. Although partial correction of anemia with erythropoiesis-stimulating agents (ESAs) results in left ventricular

hypertrophy regression, current information suggests no CV outcome benefit of normalized hemoglobin (see [Chapter 86](#)).

Secondary Hyperparathyroidism and Mineral Metabolism

Disturbances of calcium and phosphate metabolism might accelerate calcifying atherosclerosis and arteriosclerosis (see also [Chapter 88](#)). Chronically elevated FGF-23 levels are associated with high rates of left ventricular hypertrophy, atrial fibrillation, and mortality. In registry data, a strong independent mortality risk is predicted by hyperphosphatemia, an intermediate risk by elevated serum calcium levels, and a weak risk by high or low serum intact parathyroid hormone levels.¹⁷ The overall proportion of mortality risk that is potentially attributable to mineral metabolism disorders is estimated to be about 17% in HD patients.

Cardiovascular Calcification

CV calcification may affect the arterial media, atherosclerotic plaques, myocardium, and heart valves. Medial calcification causes arterial stiffness and, consequently, increased pulse pressure.¹⁸ The pathophysiologic role of plaque calcification is less clear because soft plaques are assumed to rupture and cause acute MI; atherosclerotic calcification is a potent risk marker for CV events, but its utility as a risk marker for clinical management of CKD patients remains controversial. Valvular calcification mostly affects the aortic and mitral (annulus) valves in dialysis patients and contributes to progressive stenosis and associated morbidity and mortality. In dialysis patients, extensive vascular (especially coronary artery) calcification can occur at young ages. Calciphylaxis (calcific uremic arteriolopathy) is discussed in [Chapter 91](#).

Vascular calcification is not derived only from passive calcium and phosphate precipitation. Rather, it involves differentiation of vascular smooth muscle cells toward osteoblasts induced by phosphate, calcium, and other factors, such as calcitriol and proinflammatory cytokines. Uremic bone disease, senescence, vitamin K deficiency, genomic damage, persistent inflammation, somatic mutations, posttranslational protein modification and protein-energy wasting may be additional risk factors for vascular calcification.¹⁹ One mechanism by which chronic inflammation promotes

vascular calcification may involve downregulation of fetuin-A, the most potent circulating inhibitor of extrasosseous calcification and a component of calciprotein particles. Apart from fetuin-A, other inhibitors, such as magnesium, probably counteract unwanted calcification. Leptin, matrix GLA protein, FGF-23, pyrophosphates, bone morphogenic proteins (e.g., BMP-2 and BMP-7), and osteoprotegerin may be related to accelerated vascular calcification in ESKD. Deficiency of vitamin K and/or treatment with vitamin K antagonists (warfarin) accelerate the vascular calcification process in the uremic milieu.²⁰ Recent studies using dual-energy computed tomography imaging show vascular urate crystalline deposits in the aorta and coronary arteries of a high percentage of subjects with gout, and these urate deposits sometimes colocalize with sites of vascular calcification.²¹ While studies have not investigated whether this also may occur in CKD, it also raises the possibility that uric acid may also be involved as a nidus for calcification, similar to what occurs in the kidney tract.

Advanced Glycation End-Products

Advanced glycation end-products (AGEs) accumulate in CKD as a result of nonenzymatic glycation, oxidative stress, intestinal food components, and diminished clearance of AGE precursors. These AGEs interact with the receptor for advanced glycation end-products (RAGE); a key molecule in the genesis of CV and arterial thrombosis in CKD.²² Stable AGE residues of long-lived proteins are biomarkers of cumulative metabolic, inflammatory, and oxidative stress; carbonyl stress is speculated to contribute to tissue aging and long-term CKD complications. Whether AGE inhibition may affect CV disease in CKD is unknown.

Dialysis Modality

Reports from dialysis registries are inconsistent regarding whether HD or PD is associated with better outcomes. Valid mortality comparisons between HD and PD modalities are not available because this would require stratification of patients according to underlying ESKD cause, age, and level of baseline comorbidity. Cardiac arrhythmias, such as atrial fibrillation, seem to occur more often on the day of HD compared with PD.²³

SELF-ASSESSMENT QUESTIONS

- Which of the following is *not* an established risk factor for CV disease in patients with advanced CKD?
 - Inflammation
 - Smoking
 - Dyslipidemia
 - Hyperhomocysteinemia
 - Diabetes
- What is *not* true about cardiovascular complications in patients with ESKD?
 - Left ventricular hypertrophy is present in about 75% of the patients.
 - In dialysis patients the incidence of atrial fibrillation is about 15% per year.
 - Low levels of cholesterol predict death.
 - In dialysis patients, extensive vascular, especially coronary artery, calcification can occur even at young ages.
 - The calcific aortic stenosis progression rate is only slightly higher in dialysis patients compared with the general population.
- What is the *most* common cause of death in dialysis patients?
 - Congestive heart failure
 - Infection
 - Arrhythmic death
 - Stroke
 - Valvular heart disease
- Which drug has been shown to reduce mortality in dialysis patients with dilated cardiomyopathy and systolic heart failure?
 - Atorvastatin
 - Digoxin
 - Carvedilol
 - Metoprolol
 - Atenolol

REFERENCES

- Go AS, Chertow GM, Fan D, et al. Chronic kidney disease and the risks of death, cardiovascular events, and hospitalization. *N Engl J Med*. 2004;351:1296–1305.
- Wanner C, Amann K, Shoji T. The heart and vascular system in dialysis. *Lancet*. 2016;388:76–284.
- Boenink R, Stel VS, Waldum-Grevbo BE, Collart F, Kerschbaum J, Heaf JG, et al. Data from the ERA-EDTA Registry were examined for trends in excess mortality in European adults on kidney replacement therapy. *Kidney Int*. 2020;98:999–1008.
- U.S. Renal Data System. *USRDS 2020 Annual Data Report: Atlas of Chronic Kidney Disease & End-Stage Renal Disease in the United States*. Bethesda, MD: National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases; 2020.
- Stenvinkel P, Carrero JJ, Axelsson J, Lindholm B, Heimbürger O, Massy Z. Emerging biomarkers for evaluating cardiovascular risk in the chronic kidney disease patient: how do new pieces fit into the uremic puzzle? *Clin J Am Soc Nephrol*. 2008;3:505–521.
- Eckardt KU, Gillespie IA, Kronenberg F, Richards S, Stenvinkel P, Anker SD, et al. High cardiovascular event rates occur within the first weeks of starting hemodialysis. *Kidney Int*. 2015;88:1117–1125.
- Vanholder R, Gryp T, Glorieux G. Urea and chronic kidney disease: the comeback of the century? (in uraemia research). *Nephrol Dial Transplant*. 2018;33:4–12.
- Kooman JP, Kotanko P, Schols AM, Shiels PG, Stenvinkel P. Chronic kidney disease and premature ageing. *Nat Rev Nephrol*. 2014;10:732–742.
- Rapp N, Evenepoel P, Stenvinkel P, Schurgers L. Uremic toxins and vascular calcification-missing the forest for all the trees. *Toxins (Basel)*. 2020;12:624.
- Himmelfarb J, Stenvinkel P, Ikizler T, Hakim RM. The elephant in uremia: oxidant stress as a unifying concept of cardiovascular disease in uremia. *Kidney Int*. 2002;62:1524–1538.
- Kooman JP, Dekker MJ, Usvyat LA, Kotanko P, van der Sande FM, Schalkwijk CG, et al. Inflammation and premature aging in advanced chronic kidney disease. *Am J Physiol Renal Physiol*. 2017;313:F938–F50.
- Cobo G, Quershi AR, Lindholm B, Stenvinkel P. C-Reactive protein: repeated measurements will improve dialysis patient care. *Semin Dial*. 2016;29:7–14.
- Sturmlechner I, Durik M, Sieben CJ, Baker DJ, van Deursen JM. Cellular senescence in renal ageing and disease. *Nat Rev Nephrol*. 2017;13:77–89.
- Stenvinkel P, Gillespie IA, Tunks J, et al. Inflammation modifies the paradoxical association between body mass index and mortality in hemodialysis patients. *J Am Soc Nephrol*. 2016;27:1479–1486.
- Tang WH, Hazen SL. The contributory role of gut microbiota in cardiovascular disease. *J Clin Invest*. 2014;124:4204–4211.
- Stenvinkel P, Carrero JJ, Axelsson J, et al. Emerging biomarkers for evaluating cardiovascular risk in the chronic kidney disease patient: how do new pieces fit into the uremic puzzle? *Clin J Am Soc Nephrol*. 2008;3:505–521.
- Memmos E, Papagianni A. New insights into the role of FGF-23 and Klotho in cardiovascular disease in chronic kidney disease patients. *Curr Vasc Pharmacol*. 2021;19(1):55–62.
- Nelson AJ, Raggi P, Wolf M, Gold AM, Chertow GM, Roe M. Targeting vascular calcification in chronic kidney disease. *JACC Basic Transl Sci*. 2020;5:398–412.
- Dai L, Schurgers LJ, Shiels PG, Stenvinkel P. Early vascular ageing in chronic kidney disease: impact of inflammation, vitamin K, senescence and genomic damage. *Nephrol Dial Transplant*. 2020;35(suppl 2): ii31–ii7.
- Cozzolino M, Fusaro M, Ciceri P, Gasperoni L, Cianciolo G. The role of vitamin K in vascular calcification. *Adv Chronic Kidney Dis*. 2019;26:437–444.
- Klauser AS, Halpern EJ, Strobl S, Gruber J, Feuchtnner G, Bellmann-Weiler R, et al. Dual-energy computed tomography detection of cardiovascular monosodium urate deposits in patients with gout. *JAMA Cardiol*. 2019;10:1019–1028.
- Ortillon J, Hézard N, Belmokhtar K, Kawecki C, Terryn C, Fritz G, et al. Receptor for advanced glycation end products is involved in platelet hyperactivation and arterial thrombosis during chronic kidney disease. *Thromb Haemost*. 2020;120:130–1312.
- Chang CH, Fan PC, Lin YS, Chen SW, Lin MS, Wu M, et al. Atrial fibrillation and associated outcomes in patients with peritoneal dialysis and hemodialysis: a 14-year nationwide population-based study. *J Nephrol*. 2020.